

Gas Usage Forecasting and Delivery Optimization for Scot Forge

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Motivation & Problem

Scot Forge is an employee-owned forging manufacturer whose furnaces at two Illinois sites (**Spring Grove** and **Franklin Park**) consume large volumes of natural gas daily. Recent **Illinois tariff changes** introduced a three-tier penalty structure that depends on:

- Daily injection or withdrawal relative to contract caps
- Month-end inventory relative to seasonal storage bands
- Cumulative cash-out events by penalty tier (Tier 1 – Tier 3)

A *forecast-only* solution proved insufficient: even an accurate demand forecast produces expensive decisions when it ignores storage dynamics. The project was reframed as a full **forecasting-and-optimization pipeline**.

Dataset

Property	Value
Observations	3,410 daily records
Period	Aug 2015 – Nov 2024
Usage streams	3 metered series
Storage capacity	144,841 Dth
Backtest window	Jan 2022 – Sep 2024

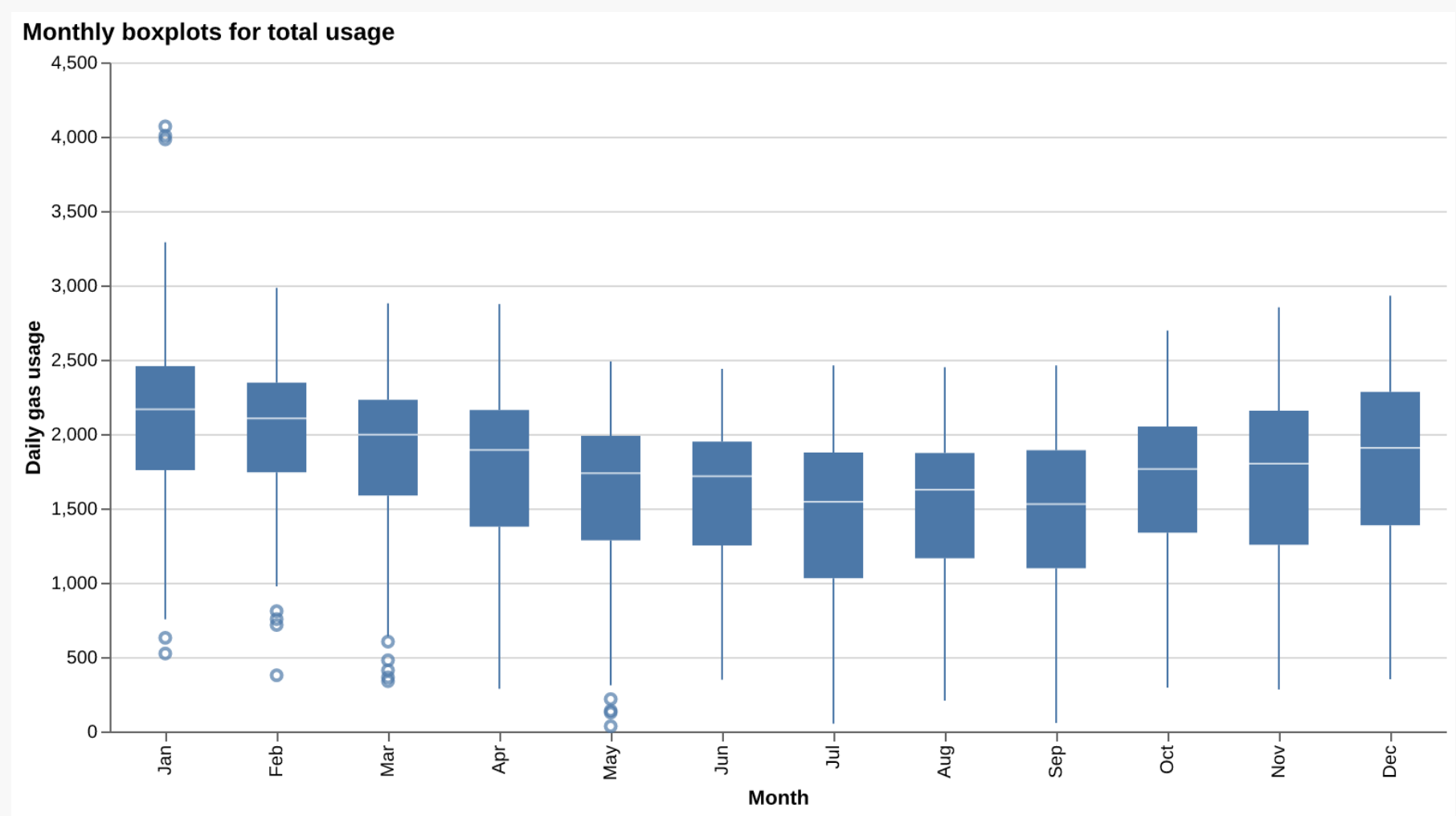
Stream composition of total demand:

- Usage 2 (Site 2, Meter 1): **84.5%** — dominant driver
- Usage 1 (Site 1): **13.6%**
- Usage 2_1 (Site 2, Meter 2): **2.0%** — persistent load

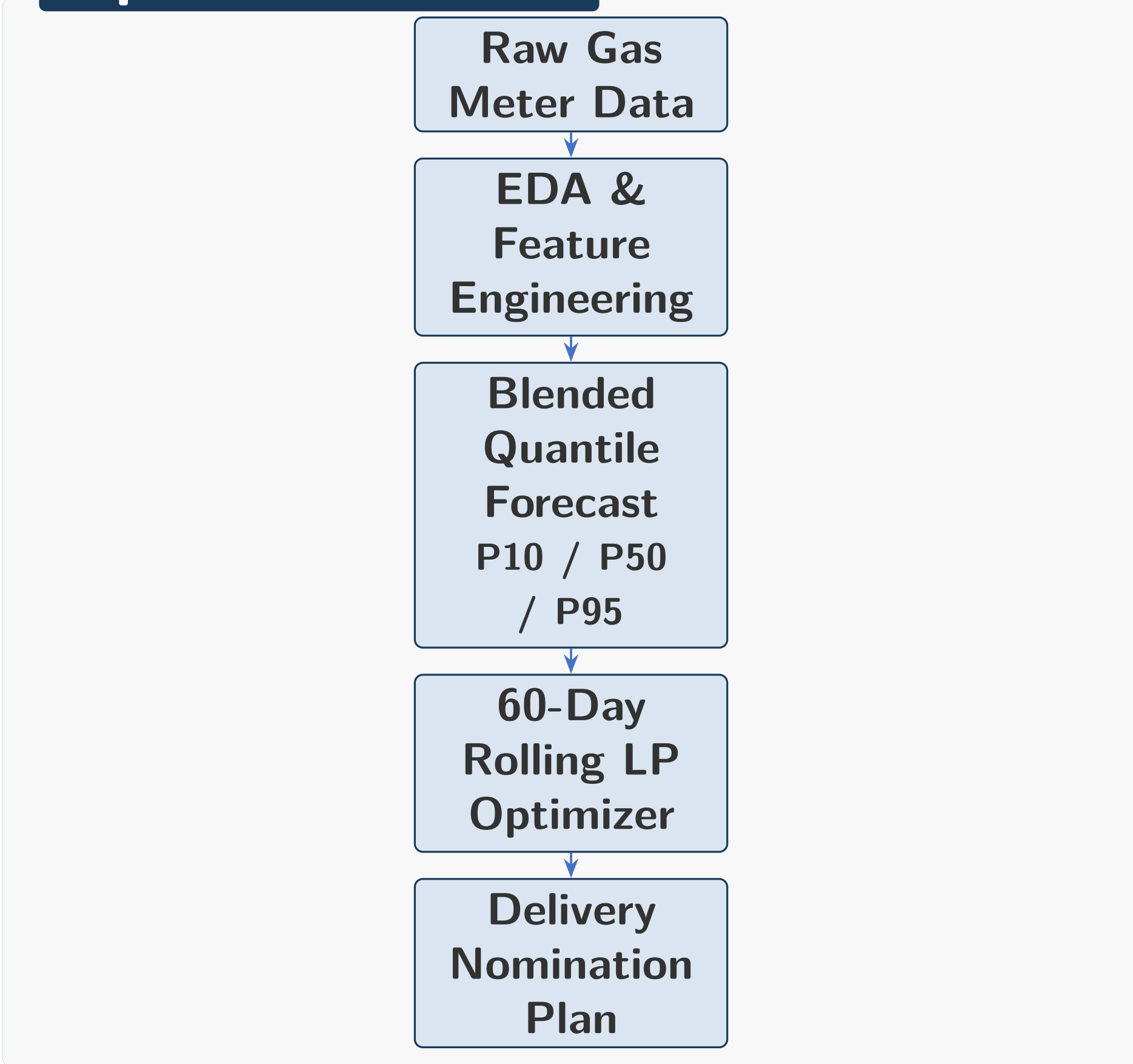
Exploratory Analysis

Key patterns identified:

- Pronounced seasonal cycle: winter peaks ($\approx 2,100$ units) vs. summer trough ($\approx 1,450$ units)
- Weekly operating cycle — lowest demand on Saturdays
- Holiday effects visible in all three streams
- 7-day lag dominates Usage 1 and Total (autocorrelation 0.82)
- Usage 2_1 highly persistent even at Lag 30 (0.808)
- Delivery–usage cumulative gap oscillates by $\pm 80,000$ Dth, confirming that storage state must be modelled explicitly



Pipeline Overview



Forecasting Model

Architecture: Blended quantile regression retrained at the start of each month. Each quantile (P10/P50/P95) blends a nonlinear *GradientBoostingRegressor* (70%) with a linear *QuantileRegressor* solved via HiGHS (30%).

Feature groups:

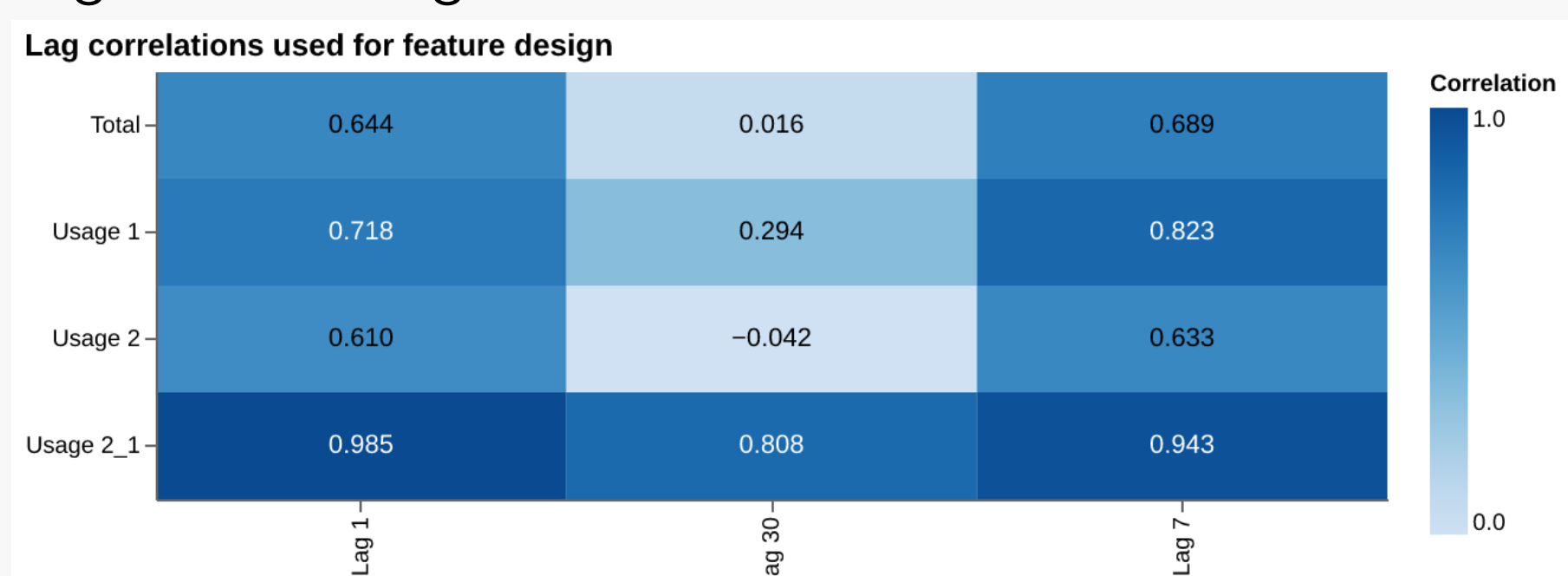
- Calendar: day-of-week, month, quarter, day-of-year
- Business: holiday flags, days-to-end-of-month
- Lags: $t-1$, $t-2$, $t-7$, $t-30$
- Smoothing: rolling mean and std at 7, 14, 30 days
- Storage: end-of-month % history

Production constraint: only information through day $t-2$ is used to forecast day t .

Benchmark results (Total Usage):

Model	RMSE	MAE	MAPE
XGBoost	270.5	203.8	17%
LightGBM	262.1	194.5	19%
CatBoost	258.6	191.8	20%
CatBoost + regime + residual	258.0	191.8	19%
TimesNet (Usage 1)	10.7	8.9	3%
PatchTST (Usage 2)	255.1	184.9	20%

Lag correlations guided feature selection:



Optimization Model

Solver `scipy.optimize.linprog(highs)`
Horizon Rolling 60-day window
Input Conservative P95 usage forecast

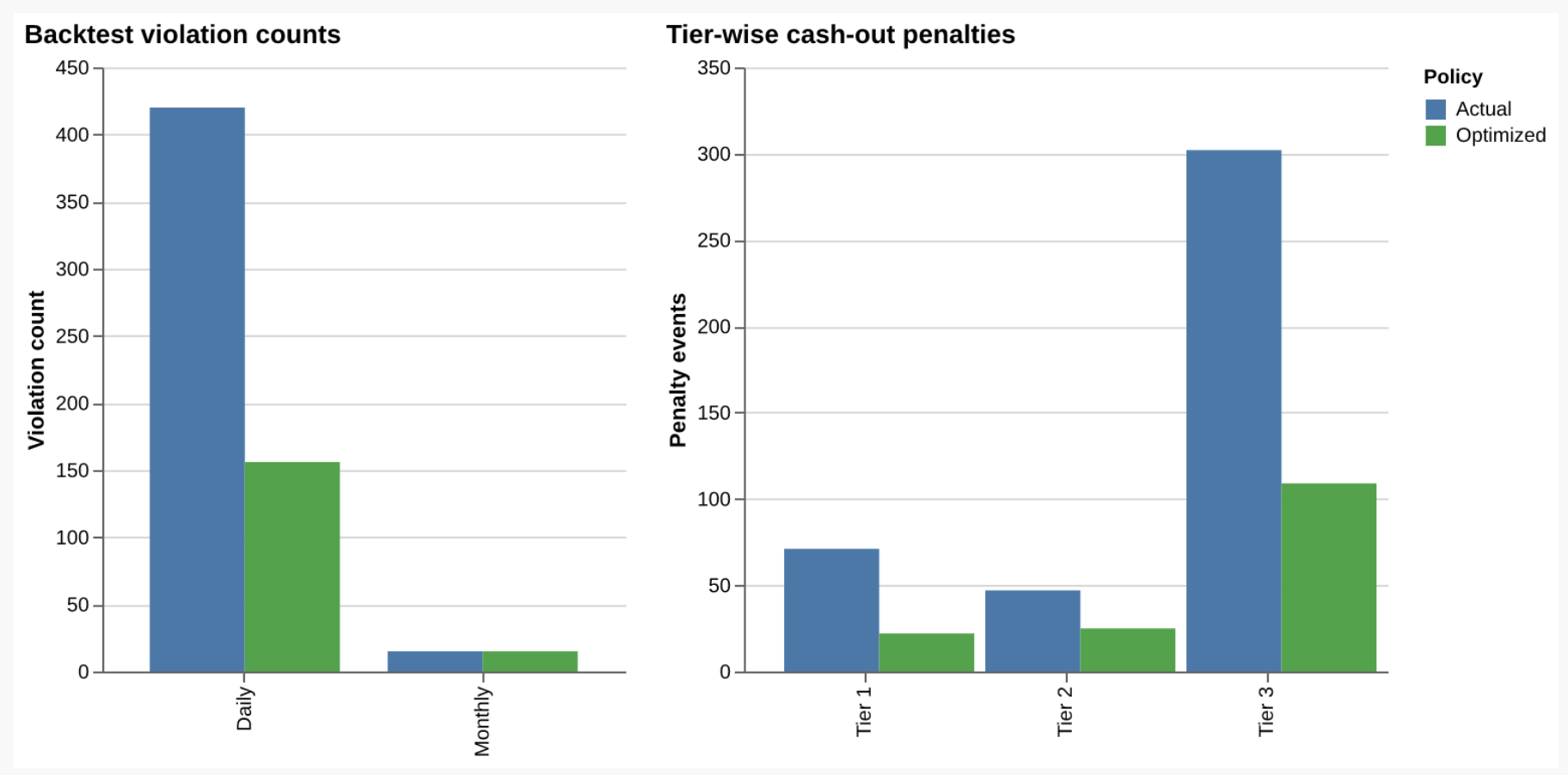
Constraints:

- Daily injection caps (month-specific % of 144,841 Dth)
- Daily withdrawal caps (month-specific)
- Monthly EOM storage bands (0–10% spring; 85–100% October)
- Mid-month steering target; adaptive seasonal buffers
- Feasibility fallback relaxes soft constraints if infeasible
- Using P95 (vs. mean) prevents the systematic under-nomination that historically caused withdrawal breaches.

Key Results — 2022–2024 Backtest

62.9%
reduction in daily penalty violations
420 → 156 events

KPI	Actual	Opt.	Δ	Reduc.
Daily violations	420	156	−264	62.9%
Monthly EOM violations	15	15	0	0.0%
Tier 1 cash-out events	71	22	−49	69.0%
Tier 2 cash-out events	47	25	−22	46.8%
Tier 3 cash-out events	302	109	−193	63.9%



Why Optimization Wins

Point-forecast accuracy (RMSE/MAE) was insufficient — tree models scored competitively yet still produced daily breaches

Quantile forecasts gave the optimizer a risk-aware usage range; P95 avoids the systematic under-nomination that triggers withdrawal violations

A **60-day horizon** lets the LP steer storage toward future EOM obligations rather than reacting to each single day

Adaptive monthly buffers addressed asymmetric injection/withdrawal risk across seasons

Monthly EOM compliance (15 events) remains unchanged — the hardest constraint even after tuning

Conclusions & Future Work

Delivered: a production-oriented pipeline that handles a two-day information lag, converts demand uncertainty into P10/P50/P95 scenarios, and chooses penalty-aware delivery nominations over a rolling 60-day horizon.

Core lesson:

The best model is not the one with the smallest error metric, but the one that produces the best *feasible decision* under business constraints.

Future directions:

- Replace violation count objective with dollar-weighted penalty cost
- Add temperature and weather forecasts as exogenous features
- Explore mixed-integer programming for discrete delivery steps
- Live deployment with automated monthly retraining and live meter feeds